

Clause & Effect: method note

Problem. Which housing rules apply to one building on one date, and what is about to change? Law is layered (state, then city), the mailing city is often not the legal city, and enacted, pending and failed law look alike in the text. Every answer must trace back to its source.

Method. The model reads, code decides.

1. **Extract.** Claude (`claude -p --model sonnet`) turns each document into schema-constrained rule records with a machine-readable `coverage` object. Prompts name no specific law. Responses are cached by `sha256(system + prompt + schema)`, so an offline rerun rebuilds `lookups.json` and `changes.json` byte for byte.
2. **Verify.** Code checks every `quoted_span` verbatim against its document. A failure gets one retry, then snaps to the closest exact passage or the record is dropped.
3. **Consolidate.** One pass per jurisdiction merges duplicates, audits coverage and turns empty cells into cited *no-rule findings*.
4. **Resolve.** The US Census Geocoder places each address in its incorporated place (486 batch, 7 single-line, 7 via OpenStreetMap). 38 of 500 addresses have a mailing city that is not their legal city.
5. **Evaluate.** Deterministic code tests time, coverage (year built, units, use) and precedence, giving `applies`, `unknown`, `superseded`, `not_yet_effective` or `pending`. 500 addresses take 0.3 s for any date.
6. **Track change.** Change tests diff two dates and list affected addresses and conflict flags. `ingest-new` adds a new law in one command.

Data and sources. The starter corpus: 87 documents for 3 states and 10 cities, 54 full texts and 33 links only. Each link was requested once, 2 s apart; 14 were readable. They keep URL and retrieval date and are flagged `secondary_source` up to the UI; the organisers confirmed this is allowed. Sites refusing automated access were skipped. No owner, resident, rent or pricing data.

Validation. `selfcheck` re-checks schema, verbatim spans, the jurisdiction × category matrix, all 25 examples in the brief, the four known open questions and the change tests. CI runs it on every pull request with the model binaries stubbed out. Output: 62 rules (57 in force, 1 not yet effective, 2 pending, 2 failed), 29 no-rule findings, every span verbatim.

Test	Result
T1 CA AB 325 / SB 763	250/250 CA addresses: <code>not_yet_effective</code> 2025-12-31, <code>applies</code> 2026-01-02
T2 Hoboken, Jersey City bans	40 + 50 addresses, each ban only in its city, none in Newark
T3 NJ FAIR Act	140/140 NJ addresses flip on 2027-07-02; 90/90 in JC and Hoboken flagged
T4 MA S.2983 / H.5222	both <code>pending</code> , never in force; 110/110 MA addresses
T5 MA ballot question	recorded as <code>failed</code> ; affected set empty

Responsible design. "Not legal advice" on every page and API response. `unknown` names the missing fact instead of guessing (626 answers). Enacted, not-yet-effective, pending and failed law stay separate. 9 rules carry a conflict flag for human review (721 answers). `audit.jsonl` logs every model call with prompt hash, model and raw output. No free-form chat, so no channel for evasion advice.

Limitations. Jersey City and Hoboken algorithmic bans rest on secondary sources. Link-only law we could not read evaluates as `unknown`. 212 addresses lack year built, 242 lack unit count; owner occupancy and subsidies are not in the data. The corpus is frozen at 2026-10-01.

Reproduce (no API key, about 5 s):

```
git clone https://github.com/isaaclins/rental-law-navigator && cd rental-law-navigator && uv sync
uv run python -m navigator run-all # from cache: extract, evaluate, changes, selfcheck
uv run python -m navigator lookup A0016 --as-of 2027-07-02
```